

Offline Adaptive Radiotherapy Implementation and Lessons Learned

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OFFLINE ADAPTIVE RADIOTHERAPY (ART)

- Comprehensive approach that leverages
 - advanced imaging
 - adaptive treatment planning techniques
 - to improve treatment precision,
 - accommodate anatomical changes with an optimized treatment regimen,
 - and ultimately maximize patient-specific treatment outcomes.
- Offline ART is a complex process that allows a treatment plan to be sufficiently adapted throughout a treatment course.
 - Thus, sophisticated workflows are very critical to ensure safe, effective, and efficient implementation of offline ART.



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OFFLINE ADAPTIVE RADIOTHERAPY (ART)

- Imaging modalities such as CBCT and MRI are commonly used to get a detailed image to detect inter-fractional variations that might occur between the planning and treatment times.
 - It is a crucial step in ART to evaluate the impact of such anatomical changes on tumor volume, shape, location, and OARs in its proximity for informed treatment plan adaptation.
- While real-time and online ART might represent the long-term vision for ART for many clinicians, offline adaptation continues to play a critical role in contemporary clinical practice.
 - Offline ART offers a resource-conscious approach to managing systematic anatomical and dosimetric changes that emerge over the course of treatment, particularly in disease sites where changes are gradual rather than acute.
 - In many institutions, offline ART serves not only as an effective clinical strategy but also as a gateway to more advanced adaptive workflows, enabling teams to build experience with imaging, deformable registration, dose accumulation, and decision-making before transitioning to online or real-time adaptation.



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Clinical Necessity of Offline Adaptive Radiotherapy

- Unlike online adaptation, offline adaptive replanning is performed outside of the treatment session, allowing more time for comprehensive plan adaptation and evaluation.
 - Thus, offline ART functions as a monitoring system that
 - leverages advanced deformable image registration (DIR) techniques, including manual or automatic error corrections for contour propagation and more accurate dose accumulation.



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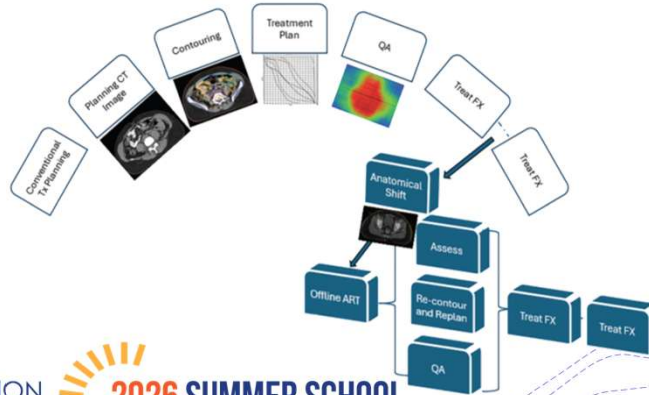


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Clinical Necessity of Offline Adaptive Radiotherapy

- For instance, if a decrease in tumor coverage or loss in organ at risk control was found due to an anatomical change,
- offline ART is to be utilized, and the revised treatment plan is used for the subsequent fractions.
 - This ensures that therapeutic goals are maintained or adapted with target dose escalation or normal tissue toxicity reduction throughout the course of radiation therapy.



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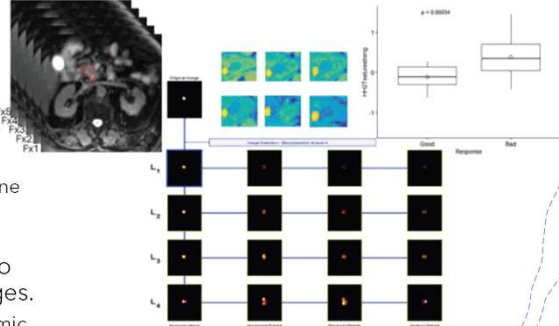
Clinical Triggers for Initiating Offline Adaptive Replanning

- There are many factors that can trigger the need to initiate an offline ART:
 - Weight loss or gain
 - Gross tumor response observed on daily images
 - Fiducial marker displacement
 - Systematic organ deformation or mass changes (e.g., bladder filling, bowel gas)
 - Deviations in delivered dose compared to planned dose (as observed through dose accumulation or DIR)

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Clinical Triggers for Initiating Offline Adaptive Replanning

- These triggers prompt the need to reassess the quality of the original plan based on the new anatomy.
- Periodic DIR and dose accumulation can be done during treatment to evaluate the deviations in delivered dose compared to the planned dose.
 - A threshold is then set in the workflow to determine the dosimetric impact and act as a cutoff for the necessity of replanning.
- Determine the temporal changes correlated to gross tumor response observed on daily images.
 - delta-radiomic analysis, high-order wavelet radiomic features maps, and structural similarity indices can be used
 - Trigger necessity of treatment adaptation to offer potentially larger organ at risk sparing while maintaining tumor control.



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Clinical Implementation

- For offline ART to be adopted in the clinical workflow,
 - multiple QA processes need to be implemented to validate the precision of the procedures and guarantee their effectiveness.
 - reduces the probability of delivering insufficient doses to the tumor or excessive doses to healthy organs because of the anatomical variations.
 - Offline ART has been used frequently for head and neck, pelvic, and thorax cancer treatments on C-arm LINACs to reoptimize the planned dose using single or multiple time points during treatment



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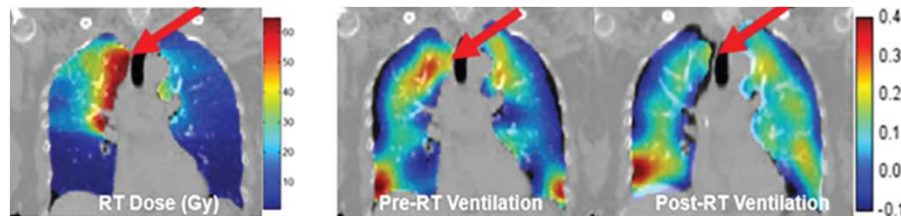


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Clinical Implementation

- Depending on the intent of adaptation, different imaging modalities may be incorporated in the replanning process.
 - When tumor shrinkage is detected on daily CBCTs,
 - gross tumor volume (GTV) can be reduced with the aid of Positron Emission Tomography (PET)/CT images,
 - which not only confirm anatomical changes but also provide biological guidance for dose escalation on residual radioresistant targets.
 - If the adaptation aims to reduce radiation toxicity to normal tissues, functional images may be incorporated. For example, ventilation images can be derived from 4DCT and used to spare functionally important lung regions during adaptive planning.



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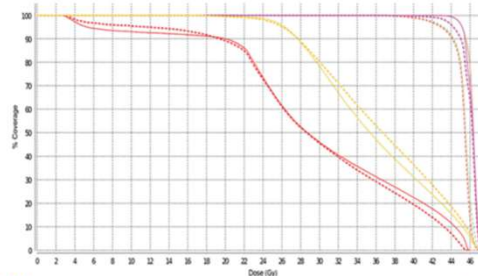
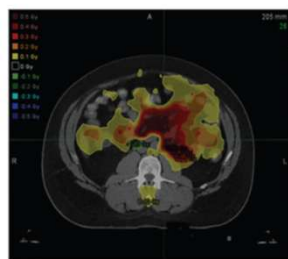
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Figure courtesy of Hualiang, Zhong

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Clinical Implementation

- To develop a linac-based workflow for offline adaptive replanning, images throughout the treatment course need to be acquired for evaluation.
- This would allow the direct comparison of the dosimetric differences between the adaptive plan and the original plan using Dose-Volume Histogram (DVH)s and dose difference evaluation.
- This automated tracking system can act as an automated watchdog that is triggered in an adaptive environment because of an anatomic change that results in a dose difference within an editing ring of >5% of the prescription dose.



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Figure courtesy of Hualiang, Zhong

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Practical Clinical Workflow Considerations for Offline Adaptive Radiotherapy

- For offline ART to be effectively done,
 - collaboration and clear communication between all the care team members are needed to effectively devise dose regimens and execute adaptive therapy strategies.
 - The radiation oncologist needs to edit the contours of the targets and critical structures within the proximity of the tumor,
 - the medical dosimetrist needs to generate a quick replan,
 - and the medical physicist needs to check the replan and perform QA of the new plan to ensure effectiveness and precision.
- In between these steps,
 - auto-segmentation or DIR techniques can be executed to transfer the contours outside of the editing ring to the replanning image.
 - Depending on the ART systems, either on-board CT/CBCT or synthetic CT images can serve as the replanning image.
 - Image quality and automatic contour QA checks,
 - image registration accuracy verification, manual editing and review for complicated structures,
 - independent dose calculations, and QA need to be developed and applied prior to delivering the adapted treatment.



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Factors Affecting the Successful Implementation of Offline Adaptive Radiotherapy

- There are other factors that need to be considered to successfully implement offline ART in a clinic:
 - Treatment workflow logistics
 - Delays and changes in treatment logistics
 - Turn around time to avoid significant treatment interruptions that can alter the treatment outcomes.
 - Extra time needed for retrospective dosimetric analysis
 - Resource limitations in smaller clinics



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Factors Affecting the Successful Implementation of Offline Adaptive Radiotherapy

- Imaging logistics
 - Availability of the CT simulator for reimaging
 - Additional time needed for extra imaging during the treatment course
 - Reliability of the image registration and dose accumulation tools
 - Imaging protocols
 - Availability of advanced imaging techniques for functional/biology-guided adaptive replanning



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Factors Affecting the Successful Implementation of Offline Adaptive Radiotherapy

- Treatment planning logistics
 - Tumor location
 - Availability of retreat workflows with appropriate dose accumulation techniques
 - Motion management techniques
 - Patient positioning reproducibility and repeatability
 - Clinical relevance of the anatomical changes on dose distribution
- Quality assurance
 - Contour QA checks
 - Replan QA checks



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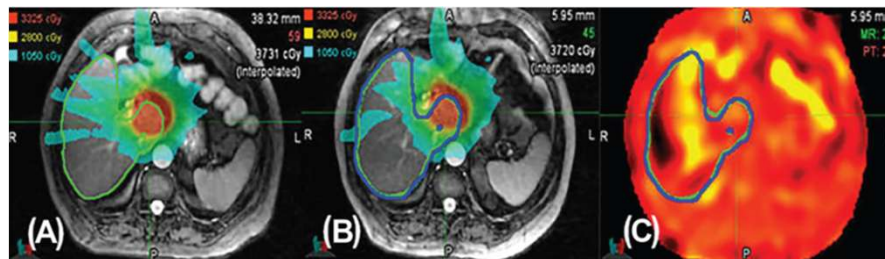
Efficient Use of Offline Adaptive Radiotherapy

- The effectiveness of offline ART relies heavily on imaging quality.
 - In clinical practice, several factors contribute to variability in the acquired image quality, which constitute one of the major drawbacks for ART implementation.
 - Image acquisition protocols,
 - patient preparation (e.g., bladder or bowel filling),
 - and reproducibility can affect the ability to detect anatomical changes, leading to either over- or underestimation of the necessity for plan adaptation.
- To address these challenges, the AAPM Task Group No. 132 that provides guidance on the use, validation, QA, and accuracy assessment of algorithms for registration and fusion of multimodality medical image datasets under a variety of clinical circumstances

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Efficient Use of Offline Adaptive Radiotherapy

- To efficiently use offline ART, we need to account for the uncertainty in image registration because accurate contour propagation between two images does not entail accurate deformable dose accumulation, as the latter relies on DIR performances at individual interior points in these organs.
 - A deformed liver contour (green) in good agreement with the target contour (blue); however, the deformed dose distribution differs substantially from the original dose
 - This discrepancy is most likely caused by displacement errors within the liver.
 - Jacobian determinants of the registration are spatially nonuniform within the liver, indicating that the registration-derived deformation map contains substantial local constriction and expansion.



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Efficient Use of Offline Adaptive Radiotherapy

- Another factor limiting the efficient use of offline ART is the lack of well-established adaptation/retreatment workflow algorithms with appropriate dose accumulation techniques to account for previously delivered doses.
- Such workflows should be capable of handling common anatomical changes including,
 - mass-conserving deformation
 - volume variation
 - mass replacement



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Lessons Learned from Clinical Implementation of Offline Adaptive Radiotherapy

- Even though, the successful implementation of offline ART relies on the availability of advanced imaging technologies and sophisticated treatment planning software.
 - such resources are not universally available across all treatment centers
 - due to their cost and the availability of sufficient highly trained multidisciplinary team to run the adaptive workflows, perform accurate image registration, interpret the results, and assess the clinical impact of the adapted plan.
 - There is also a need for continuous professional development and institutional support to maintain high-quality ART programs.



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Lessons Learned from Clinical Implementation of Offline Adaptive Radiotherapy

- Patient factors can also influence the feasibility and success of offline ART.
 - For instance, inconsistent patient compliance with preparation protocols (e.g., fasting, bladder filling),
 - changes in patient physical condition,
 - or poor performance status can affect imaging consistency and setup reproducibility.
- In some cases, the patient may not tolerate additional imaging sessions or extended planning timelines.
- Some patients may also face logistical challenges, like transport issues, that limit their availability for additional appointments.



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Lessons Learned from Clinical Implementation of Offline Adaptive Radiotherapy

- Some patients may also face logistical challenges, like transport issues, that limit their availability for additional appointments.
- These challenges must be considered carefully while developing workflows for offline ART planning tailored to the specific capabilities of the clinic and patient population.
- Offline ART workflows must include robust quality assurance processes and expert oversight, particularly when cumulative dose metrics are used to inform clinical decisions.
- Overreliance on automated tools without appropriate validation can undermine confidence in the adaptive process.



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Acceptable Timeline for Adaptive Radiotherapy

- Another limitation that can hinder offline ART success is the inherent time lag between the detection of an anatomical shift and the implementation of the adapted plan.
 - This is due to the nature of offline ART, where adaptation is done offline in between treatment sessions as opposed to online ART.
 - The need for repeat imaging, contouring of new datasets, and plan evaluation increases the workload on the team members.
 - In busy clinical settings with limited staffing and equipment availability, this can lead to additional delays in adaptation,
 - rendering the new plan less effective or even clinically irrelevant if the anatomical change that triggered it is no longer present or adequate.
 - To maintain the clinical value of offline ART, it is generally recommended that the adapted plan be implemented within two to three days of detecting the anatomical change to ensure the effectiveness of the new treatment plan and improve the treatment precision.



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Role of Interdisciplinary Communication on Successful Offline Adaptive Radiotherapy Implementation

- Offline ART represents a transformative approach in the field of radiation oncology,
 - allowing a dynamic approach to continuously monitor and evaluate the precision and effectiveness of the treatment plan offering opportunities to adapt dosimetric regimens for better treatment outcomes.
 - Successful implementation requires careful evaluation by the entire clinical team.
 - Interdisciplinary communication is critical to timely identify significant anatomical changes and to align the goals of the treatment.
 - Knowing the patient-specific challenges ahead of time can also allow the team to allocate the appropriate resources and avoid time lags between the discovery of anatomical changes and implementation of the adaptive plan.
 - Establishing a clear communication structure, shared responsibilities, and a collaborative culture enhances the quality of offline ART.
 - This allows appropriate setting of thresholds that trigger the necessity of adaptive replanning using a streamlined and well-integrated workflow.
 - Standardized procedures that are tailored to the capabilities of the clinical team and the specific needs of the patient are crucial steps to successfully implement treatment adaptation workflows that balance efficiency with demand.



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Conclusions

- Offline ART offers a structured and clinically practical approach to manage inter-fraction anatomical and physiological changes '
 - is more relevant in tumor sites with high anatomical or responsive variability in the head and neck, thoracic, and pelvic regions.
 - The successful use of offline ART is contingent upon careful patient selection, establishment of a sophisticated, well-tailored workflows that adapt to the clinic's needs, and access to high-quality imaging and planning tools.
 - Applying offline ART to patients who would benefit from treatment adaptation can enhance the precision of dose delivery, improve target coverage, and reduce unnecessary radiation exposure to OARs while maintaining an efficient use of resources.
 - The value of offline ART is heavily dependent on the balance between the clinical need and the effectiveness of interdisciplinary communication, which ensures that anatomical changes are promptly identified, appropriately evaluated, and acted upon in a timely manner.



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